Econometrics Part 3

The linear model: tests

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Outline

1. Tests
   - Reminder: common distributions
   - Tests in finite sample: the T-test
   - Tests in large sample: the T-test
2. Size, p-value, forecasting
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Usual distributions to know

- Normal
- χ^2
- Student (T)
- Fisher

The Normal distribution

It is the usual "bell-shaped" distribution, also called the Gaussian distribution.
$X \hookrightarrow N(\mu, \sigma^2)$ if its density is:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Any linear combination of independent normal random variables is a normal random variable.

The Chi-squared distribution

It is defined as the sum of squared independent standard normal variables.
$Y \hookrightarrow \chi^2_n$ if:

$$Y = \sum_{i=1}^{n} X_i^2$$

Where $X_i \hookrightarrow N(0,1)$ and the X_i are independent.
The sum of independent χ^2 variables is a χ^2: if $Y_1 \hookrightarrow \chi^2_{n_1}$ and $Y_2 \hookrightarrow \chi^2_{n_2}$ are independent, then $Y_1 + Y_2 \hookrightarrow \chi^2_{n_1+n_2}$ (but a general linear combination, e.g. $2Y_1$, is not a χ^2).

The Student distribution

It is defined as the ratio of a standard normal and the square root of an independent χ^2 divided by its degrees of freedom.
$Y \hookrightarrow T_n$ if:

$$Y = \frac{X}{\sqrt{\frac{Z}{n}}}$$

Where $X \hookrightarrow N(0,1)$ and $Z \hookrightarrow \chi^2_n$, and are independent. The Student distribution is bell-shaped with fatter tails than the normal, and converges to $N(0,1)$ when $n \to +\infty$.

The Fisher distribution

It is defined as the ratio of two independent χ^2, each divided by its degrees of freedom.
$Y \hookrightarrow F_{p,q}$ if:

$$Y = \frac{Z_1/p}{Z_2/q}$$

Where $Z_1 \hookrightarrow \chi^2_p$ and $Z_2 \hookrightarrow \chi^2_q$, and are independent.

Reminder: OLS estimators are BLUE

We still consider that the X variables are not random at this stage.

- We have $\hat{b} = (X'X)^{-1}X'y$
- $E(\hat{b}) = b$: unbiased
- $V(\hat{b}) = \sigma^2(X'X)^{-1}$, the smallest among linear unbiased estimators
- $\hat{\sigma}^2 = (\sum \hat{u}_i^2)/(N-k)$: unbiased
- So $\hat{V}(\hat{b}) = \hat{\sigma}^2(X'X)^{-1}$
- Gauss-Markov theorem: under hypotheses 1 to 4, the OLS estimator is the Best Linear Unbiased Estimator (unbiased with a minimal variance)
- So far, no assumption on the distribution of variables (y, X or u)
- So we don't know yet the distribution of the estimator $\hat{b}$, which is needed for tests

Distribution of estimators in a finite sample (1)

- Finite sample = sample that is not necessarily large
- We need to assume that $u \hookrightarrow N(0, \sigma^2 I_N)$
- Then, since $\hat{b} = b + (X'X)^{-1}X'u$ is linear in u: $\hat{b} \hookrightarrow N(b, \sigma^2(X'X)^{-1})$
- Calling $\hat{b}_j$ the jth element of $\hat{b}$ and α_{jj} the jth element of the diagonal of $(X'X)^{-1}$, $\sigma^2_{\hat{b}_j} = \sigma^2 \alpha_{jj}$:
- $\hat{b}_j \hookrightarrow N(b_j, \sigma^2_{\hat{b}_j})$
- Next, we standardize $\hat{b}_j$ to have it follow a standard Normal distribution:

$$\forall j, \frac{\hat{b}_j - b_j}{\sigma_{\hat{b}_j}} \hookrightarrow N(0,1)$$

Distribution of estimators in a finite sample (2)

- The previous expression could be used to test possible values of b_j, given $\hat{b}_j$ and $\sigma_{\hat{b}_j}$
- But $\sigma_{\hat{b}_j}$ is unknown: it depends on σ, the standard deviation of the error terms
- Plugging in the estimate $\hat{\sigma}_{\hat{b}_j}$ instead of the true $\sigma_{\hat{b}_j}$ changes the distribution from a Normal to a Student
- Why: $\hat{\sigma}_{\hat{b}_j}$ is an imperfect (random) estimate of $\sigma_{\hat{b}_j}$

$$\forall j, t_j = \frac{\hat{b}_j - b_j}{\hat{\sigma}_{\hat{b}_j}} \hookrightarrow T_{N-k}$$

Where $\hat{\sigma}^2_{\hat{b}_j} = \hat{\sigma}^2 \alpha_{jj}$ (cf. *supra*).
Notice that when $N \to \infty$, T_{N-k} converges to $N(0,1)$.

Formal proof

We know that $\hat{\sigma}^2 = \frac{\sum \hat{u}_i^2}{N-k}$. We can show that:

$$(N-k)\frac{\hat{\sigma}^2}{\sigma^2} \hookrightarrow \chi^2_{N-k}$$

Proof: using the fact that u is normal and that $\hat{u} = M_X u$ (see Dormont). Moreover, $\hat{\sigma}^2$ is independent of $\hat{b}$.
Then, by definition of the Student distribution (a standard normal over the square root of an independent χ^2 divided by its degrees of freedom):

$$t_j = \frac{(\hat{b}_j - b_j)/\sigma_{\hat{b}_j}}{\sqrt{\frac{(N-k)\hat{\sigma}^2/\sigma^2}{N-k}}} = \frac{\hat{b}_j - b_j}{\hat{\sigma}\sqrt{\alpha_{jj}}} = \frac{\hat{b}_j - b_j}{\hat{\sigma}_{\hat{b}_j}} \hookrightarrow T_{N-k}$$

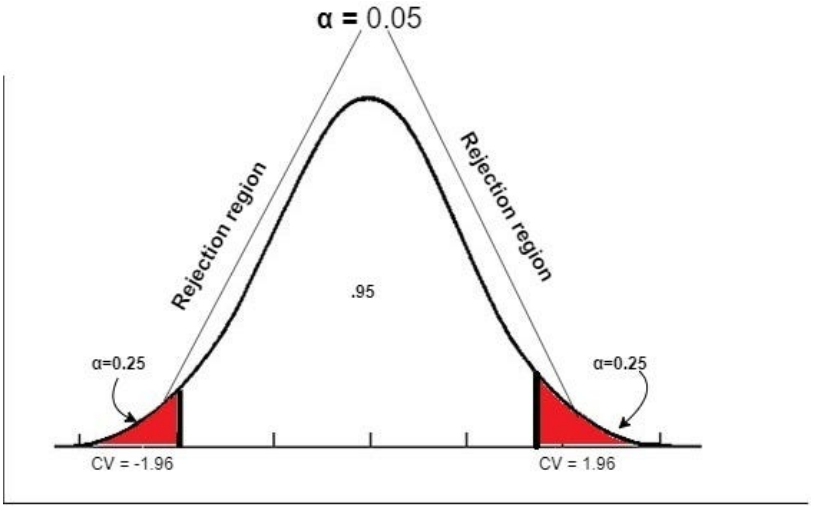
The test procedure: an example (1)

- Consider the model income $= a + b \cdot \text{height} + c \cdot \text{education} + u$, estimated over N individuals
- Suppose that the estimated parameter $\hat{b}$ is *close* to zero
- We may infer that *height* is irrelevant: its effect on income (holding education fixed) could be zero
- That is, the "true" b could be zero
- But even if it is, it is very unlikely that we get exactly $\hat{b} = 0$ (due to intrinsic randomness)
- Given the computed $\hat{b}$, we need a way to assess whether the "true" b is zero or not

The test procedure: an example (2)

- Let's call H_0 the hypothesis $b = 0$ (the *null hypothesis*), and H_1 the hypothesis $b \neq 0$
- Should we consider H_0 as true?
- We know that for this model, $t = (\hat{b} - b)/\hat{\sigma}_{\hat{b}} \hookrightarrow T_{N-3}$ ($k = 3$ parameters)
- Is the latter still likely, if we assume H_0 is true?
- Assuming H_0 is true means that $b = 0$, so that t simplifies to $t = \hat{b}/\hat{\sigma}_{\hat{b}}$
- If this value is unlikely to come from a T_{N-3} distribution, we conclude that H_0 is wrong: we *reject* it
- Rejecting $H_0 \Leftrightarrow$ parameter b is significant (significantly different from 0)
- Not rejecting $H_0 \Leftrightarrow$ parameter b is not significant

Illustration



A general test procedure

- $H_0 : b = 0$
- $H_1 : b \neq 0$

1. Find the value $N - k$
2. Compute $t = \hat{b}/\hat{\sigma}_{\hat{b}}$
3. Compare $|t|$ with the critical value t^*: the 97.5% quantile of a T_{N-k} distribution
4. We take 97.5% because the test is two-sided: we reject if t falls in the 5% least likely values of the distribution, i.e. the lowest 2.5% or the highest 2.5%
5. If $|t| > t^*$, H_0 is rejected: b is significantly different from 0 (significant)
6. Otherwise, H_0 is not rejected: b is not significantly different from 0
7. Always mention the level of significance: here (and it is often the case): $\alpha = 5\%$
8. If a variable is not significant, we may consider removing it (but see *power* below)

The larger $|t|$, the stronger the evidence against H_0.

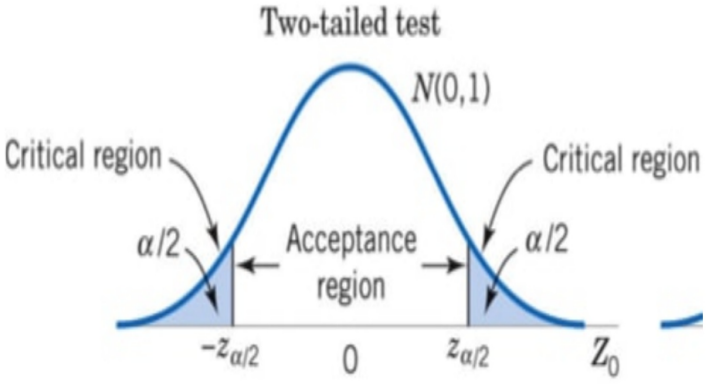
An even more general test procedure

- $H_0 : b = a$
- $H_1 : b \neq a$

1. Find the value $N - k$
2. Compute $t = \frac{\hat{b} - a}{\hat{\sigma}_{\hat{b}}}$
3. Compare $|t|$ with t^*, the 97.5% quantile of a T_{N-k} distribution
4. If $|t| > t^*$, H_0 is rejected: parameter b is significantly different from a
5. Otherwise, H_0 is not rejected: parameter b is not significantly different from a
6. Always mention the level of significance: here (and it is often the case): $\alpha = 5\%$

The larger $|t|$, the stronger the evidence against H_0.

Illustration



Is it very different?

- If $N \to \infty$, all we've covered so far still works, because it works for *any* N
- But it relies on the assumption that u is normal
- In large samples, we can drop this assumption and directly use large-sample tests
- These are called *asymptotic tests*

Consistency of estimators

We want to know what happens if the sample size becomes very large, i.e. when $N \to \infty$. Call $\hat{b}_N$ the estimate computed on a sample of size N.

- $\hat{b}_N$ is consistent: $\text{plim}(\hat{b}_N) = b$
- Proof:
- Consistency in quadratic mean implies consistency in probability
- Since $\hat{b}_N$ is unbiased, $E[(\hat{b}_N - b)(\hat{b}_N - b)'] = V(\hat{b}_N)$: does it tend to 0?
- $V(\hat{b}_N) = \sigma^2(X'X)^{-1} = \frac{\sigma^2}{N}\left(\frac{X'X}{N}\right)^{-1}$
- Assuming $\frac{X'X}{N} \to V_X$ (finite and invertible), the limit is $0 \cdot V_X^{-1} = 0$

So we see a first advantage of working with large samples: estimates are *consistent*.

Distribution of estimators

Estimators exhibit *asymptotic normality*: when $N \to \infty$, we know that their distribution is a Normal, whatever the distribution of u's (contrary to the finite sample case).
Thanks to the Central Limit Theorem (CLT), we can derive the distribution of estimators when $N \to \infty$. Call $\hat{b}_N$ the estimate computed on a sample of size N.

- $\hat{b}_N$ is asymptotically normal: $\sqrt{N}(\hat{b}_N - b) \xrightarrow{d} N(0, \sigma^2 V_X^{-1})$
- σ^2 is the variance of error terms u
- "Inflating" by $\sqrt{N}$ is needed, otherwise the limit would be degenerate (zero variance)
- Intuition:
- $\hat{b}_N - b = (X'X)^{-1}X'u$, so that it is a linear combination of the u's
- So the CLT applies and gives a normal distribution when $N \to \infty$

Test statistic

- In a large sample, $\hat{\sigma}_{\hat{b}_j}$ is an almost perfect estimate of $\sigma_{\hat{b}_j}$
- Formally, plim $\frac{\hat{\sigma}_{\hat{b}_j}}{\sigma_{\hat{b}_j}} = 1$
- Plugging in $\hat{\sigma}_{\hat{b}_j}$ instead of $\sigma_{\hat{b}_j}$ does not change the limiting distribution, it stays a Normal

$$\forall j, t_j = \frac{\hat{b}_j - b_j}{\hat{\sigma}_{\hat{b}_j}} \xrightarrow{d} N(0,1)$$

So tests are run directly using the Normal quantiles. This is consistent with running finite-sample tests on large samples, since T_{N-k} converges to $N(0,1)$ when $N \to \infty$.
Notice that statsmodels displays the finite-sample statistics (t and Student p-values) by default.

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Size and power of a test

- α = Type I error = $P(H_0$ rejected $\mid H_0$ is true)
- β = Type II error = $P(H_0$ not rejected $\mid H_1$ is true)
- α is the *size* of a test: when H_0 is true, we wrongly reject it $100\alpha\%$ of the time
- α is also called the *significance level*
- $1-\beta$ is the *power* of a test: it indicates how good a test is at detecting deviations from the null hypothesis H_0
- $1-\beta = P(H_0$ rejected $\mid H_1$ is true)
- Lowering $\alpha \Rightarrow$ increasing β (and reciprocally): there is a trade-off between the two
- Since we cannot minimize both, we set α (e.g. 5%) and look for the test that minimizes β for this given α
- Minimizing $\beta \Leftrightarrow$ maximizing the power of a test

Sample size

- Dropping a useful variable can lead to inconsistent estimates (omitted variable bias), while keeping an unimportant variable only leads to a loss in precision
- Say we have a small sample: estimates then have a large variance
- If we also set a stringent $\alpha = 0.01$, then even if the true parameter is not zero, its t statistic is likely to be too small to reject H_0, although H_0 is false

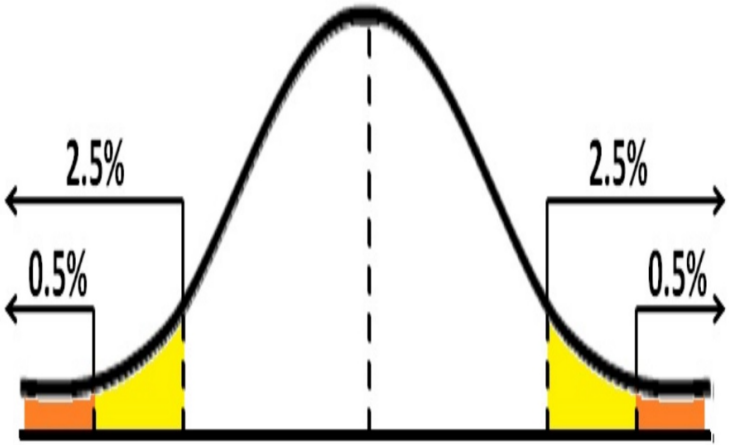
Choosing the right test size

- In that case, we might remove a relevant variable from the analysis simply because we have been too stringent about the size of the test
- Since it is easier to control α than β, we usually set α according to the sample size, in order to have enough power (i.e. ability to discriminate H_0 from the true value of the parameter)

Example

- Suppose we test the hypothesis $b = 0$, while the true value is $b = 0.1$
- The probability that we reject H_0 depends on the standard error of $\hat{b}$, thus on the sample size
- The larger the sample, the smaller the standard error, so the more likely we are to reject H_0
- Type II errors thus become increasingly unlikely as the sample size increases
- In a big sample, we can thus decrease the size of the test α: e.g. 1%
- Similarly, we can choose a size of 10% in small samples

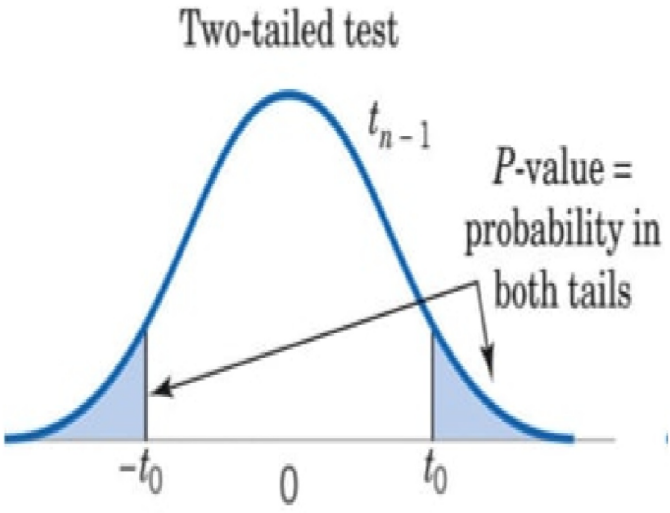
Illustration



The p-value

- It is defined with respect to the test statistic t that we just computed
- Its definition is: p-value $= P(|X| > |t|)$ with $X \hookrightarrow T_{N-k}$
- It tells where t is located in the Student distribution T_{N-k} (more precisely, at which quantile)
- Equivalently, it is the smallest size α at which H_0 would be rejected
- If p-value < 0.05, it means that $|t| > t^*$ (the 97.5% quantile)
- It means that H_0 is rejected at the 5% level
- statsmodels OLS results provide p-values for all these tests, so that decision is made easy

Illustration



Remark

- Say we are testing $b = 0$, i.e. no impact of some variable on the outcome
- If we don't reject H_0, it simply means that there is not enough evidence to reject H_0
- But another study could possibly find enough evidence to reject it
- This is why we say that H_0 is "not rejected", rather than "accepted"
- Also, the p-value is **not** the probability that H_0 is true: it is computed *assuming* H_0 is true

Using the model to make a forecast

- Using the model $y = Xb + u$ estimated on observations $1, \ldots, N$, how could we predict y_T for a new observation with regressors X_T?
- A natural prediction is: $\hat{y}_T = X_T' \hat{b}$
- Notice that X_T' is a row vector, since in the matrix of observations, one row = one observation
- And the forecast error is: $\hat{e} = y_T - \hat{y}_T$

The forecast error

- The "true" y_T is: $y_T = X_T' b + u_T$
- The forecast error is thus: $\hat{e} = X_T'(b - \hat{b}) + u_T$
- u_T is uncorrelated with the $(u_i)_{i \in [1,N]}$ and with the X's, so:
- $E(\hat{e}) = 0$: the forecast is unbiased
- $V(\hat{e}) = E(\hat{e}^2) = \sigma^2 X_T'(X'X)^{-1} X_T + \sigma^2$
- It thus depends on:
  - the variance of the error term σ^2 (noise that no model can predict)
  - the accuracy of $\hat{b}$ (its variance $\sigma^2(X'X)^{-1}$) and the distance between X_T and the X's used for the estimation

The forecast interval

The 95% forecast (prediction) interval is:

$$\hat{y}_T \pm t^* \sqrt{\hat{\sigma}^2 \left(1 + X_T'(X'X)^{-1}X_T\right)}$$

With t^* the 97.5% quantile of the T_{N-k} distribution (if N is large, it is approximately the 97.5% quantile of the normal distribution: 1.96).

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Testing a linear combination of parameters

- A T-test can be used to test a linear restriction of parameters, e.g. $b_1 + 2b_2 = a$
- For more than one linear restriction, we need the Fisher test (see below)
- $\hat{b}_1 + 2\hat{b}_2$ is normal (linear combination of jointly normal estimators), so the standardized statistic follows a Student distribution
- The test statistic is: $t = \frac{(\hat{b}_1 + 2\hat{b}_2) - a}{\hat{\sigma}_{\hat{b}_1 + 2\hat{b}_2}} \hookrightarrow T_{N-k}$ under H_0
- With $\hat{\sigma}_{\hat{b}_1 + 2\hat{b}_2}$ computed as the usual standard error of a sum of variables
- Hint: $V(\lambda X + \mu Y) = \lambda^2 V(X) + \mu^2 V(Y) + 2\lambda\mu \operatorname{cov}(X, Y)$
- Here: $\hat{V}(\hat{b}_1 + 2\hat{b}_2) = \hat{V}(\hat{b}_1) + 4\hat{V}(\hat{b}_2) + 4\,\widehat{\operatorname{cov}}(\hat{b}_1, \hat{b}_2)$

Example: Cobb Douglas function (1)

Say we use a Cobb Douglas production function to describe the technology of N different firms. y is the output, K and L the inputs (capital and labor respectively). α and β are the output elasticities of K and L respectively. A_i is the total factor productivity of each firm (comprises firm-specific efficiency). Our goal is to estimate α and β. For each firm i:

$$y_i = A_i K_i^{\alpha} L_i^{\beta}$$

Taking the log on both sides (all variables are assumed strictly positive), we get:

$$\log(y_i) = \log(A_i) + \alpha \log(K_i) + \beta \log(L_i)$$

Example: Cobb Douglas function (2)

Call $\log(A) = E[\log(A_i)]$ the average (log) total factor productivity. We can then write:

$$\log(y_i) = \log(A_i) - \log(A) + \log(A) + \alpha \log(K_i) + \beta \log(L_i)$$

And calling $u_i = \log(A_i) - \log(A)$, we have $E(u_i) = 0$ and:

$$\log(y_i) = \log(A) + \alpha \log(K_i) + \beta \log(L_i) + u_i$$

Which amounts to a classical linear model of the form $y = a + bx + cz + u$.
Note: defining $A = E(A_i)$ instead would not give $E(u_i) = 0$, since $E[\log(A_i)] \neq \log E(A_i)$ (Jensen's inequality).

Example: Cobb Douglas function (3)

- Data taken from Greene, on 27 firms
- Estimation of a Cobb-Douglas production function
- Parameters of interest: α and β, the output elasticities (not to be confused with the size and Type II error of a test!)
- If $\alpha = \beta \Rightarrow$ same elasticity for both factors
- If $\alpha + \beta = 1 \Rightarrow$ constant returns to scale

You can use the T-test to test for a single linear combination. For any number of linear combinations (including one), use the Fisher test, see below.

One-tailed tests

- We saw we could test $b = a$ for any value a: the procedure is not restricted to testing $b = 0$
- By default, software reports the $b = 0$ test, but we can ask for any test we like
- So far, we implemented *two-sided tests*: $H_0 : b = a$ vs. $H_1 : b \neq a$ (i.e. $b > a$ or $b < a$)
- We can run *one-sided tests* if relevant: e.g. $b = 0$ vs. $b > 0$, or $b = a$ vs. $b > a$
- One-tailed tests are used if the results are interesting only if they turn out in one direction
- In practice, we use only one tail of the test distribution
- One-tailed tests still posit $H_0 : b = a$, because the t statistic must be computed under a simple hypothesis H_0 (assuming $H_0 : b < a$ would be too vague)

One-tailed tests: procedure

- Say we wish to compare parameter b to value a
- While testing $H_0 : b = a$ in the two-tailed test procedure, we get p-value $= x$
- One-tailed tests still posit $H_0 : b = a$, while H_1 can either be $b > a$ or $b < a$
- If $\hat{b} > a$, then:
  - $H_1 : b > a$ leads to p-value $= \frac{x}{2}$
  - $H_1 : b < a$ leads to p-value $= 1 - \frac{x}{2}$
- If $\hat{b} < a$, then:
  - $H_1 : b > a$ leads to p-value $= 1 - \frac{x}{2}$
  - $H_1 : b < a$ leads to p-value $= \frac{x}{2}$

All this can be easily generalized to a linear restriction involving 2 parameters, e.g. $b_1 + b_2 > 0$

One-tailed tests: example

- Say we run an estimation on a large sample, and we want to compare parameter b to value 4
- $\hat{b} = 3.8$ and $\hat{\sigma}_{\hat{b}} = 0.1$
- The Student t-value is: $\frac{3.8-4}{0.1} = -2$
- Testing $H_0 : b = 4$ with the two-tailed procedure, we get p-value $= 0.046 = 4.6\%$
- Since $\hat{b} < 4$, then:
  - Testing $H_1 : b > 4$ leads to p-value $= 1 - \frac{0.046}{2} = 0.977 = 97.7\%$
  - Testing $H_1 : b < 4$ leads to p-value $= \frac{0.046}{2} = 0.023 = 2.3\%$
- So b is not significantly greater than 4, but it is significantly lower than 4 (at the 5% level)

Simultaneous testing

- We have $\hat{b} \hookrightarrow N(b, \sigma^2(X'X)^{-1})$
- Thus: $\hat{b} - b \hookrightarrow N(0, \sigma^2(X'X)^{-1})$
- Taking the quadratic form of these, "divided" by their variance matrix, we get a χ^2_k:
- $(\hat{b} - b)'[\sigma^2(X'X)^{-1}]^{-1}(\hat{b} - b) = \frac{(\hat{b}-b)'(X'X)(\hat{b}-b)}{\sigma^2} \hookrightarrow \chi^2_k$
- But σ is unknown
- In a large sample, no problem: we plug in the estimate $\hat{\sigma}$, and it is still (asymptotically) a χ^2_k

Finite vs. large sample

- In a finite sample, replacing σ by its estimate $\hat{\sigma}$ changes the distribution from a χ^2_k to a Fisher $F_{(k,N-k)}$ (after dividing by k)
- Formally: we use $(N-k)\frac{\hat{\sigma}^2}{\sigma^2}$ which is a χ^2_{N-k}, independent of $\hat{b}$
- We compute the ratio of the two χ^2, each divided by its degrees of freedom, and get a Fisher distribution

F-test continued

$$f = \frac{(\hat{b} - b)'(X'X)(\hat{b} - b)}{k\hat{\sigma}^2} \hookrightarrow F_{(k,N-k)}$$

- The testing procedure is the same as for the T-test, e.g. for $H_0 : b = b_0$:
- Assume H_0 is true, and replace b by b_0 in f
- Compute f
- Check whether it is likely that f comes from a $F_{(k,N-k)}$ distribution
- If f is larger than the 95% quantile of $F_{(k,N-k)}$, H_0 is rejected at the 5% level; otherwise, it is not rejected (the F-test only uses the upper tail)

Remark on a single-parameter Fisher test

- For a test of a single linear combination after a regression, we would normally use the Student test
- Instead, we could also use a Fisher test
- Fisher tests can be used to test any number of linear constraints, so they can be used even if there is only one
- The square of a Student variable follows a Fisher distribution, so Student and Fisher procedures are equivalent here:
- $Y \hookrightarrow T_n$ if: $Y = \frac{X}{\sqrt{Z/n}}$ with $X \hookrightarrow N(0,1)$ and $Z \hookrightarrow \chi^2_n$ independent
- Hence $Y^2 = \frac{X^2/1}{Z/n}$, with $X^2 \hookrightarrow \chi^2_1$
- This is the definition of a Fisher distribution: $Y^2 \hookrightarrow F_{1,n}$
- So $t^2 \hookrightarrow F_{1,N-k}$: the F-test of one constraint gives exactly the same p-value as the two-sided T-test

About simultaneous testing

- A F-test is about testing a group of parameters (here k parameters)
- Why couldn't we use k T-tests instead?
- First, the global F-test takes into account possible correlation between parameters
- Second, even assuming that the estimators are independent (which rarely happens), consider the following example:
- 20 parameters b_i, each tested with a T-test of size $a = 0.05$: $h_0^i : b_i = 0$ vs. $h_1^i : b_i \neq 0$
- Global test: H_0: all the parameters are equal to 0 vs. H_1: at least one of them is not, rejecting H_0 as soon as one h_0^i is rejected
- What is the actual size of this global test?
- $\alpha = P(H_0 \text{ rejected} \mid H_0 \text{ is true}) = 1 - P(\text{no } h_0^i \text{ is rejected} \mid \text{they are all true})$
- $\alpha = 1 - (1 - a)^{20} = 1 - (1 - 0.05)^{20} \approx 0.64$, far above 5%!

A generalization of the F-test (1)

- We may want to test only a subset of parameters. In that case, we simply take the preceding formula restricted to the q parameters in question:
- $f = (\hat{\beta} - \beta)'[Z/(q\hat{\sigma}^2)](\hat{\beta} - \beta) \hookrightarrow F_{(q,N-k)}$
- β: subset of parameters tested
- q: number of parameters tested (= no. of constraints)
- Z^{-1}: submatrix of $(X'X)^{-1}$ corresponding to the q coefficients
- The test that all parameters are 0 (except the constant term) is always reported by statsmodels: it is a general test of the model, called the global F-test

A generalization of the F-test (2)

- We may want to test some linear constraints (ex: $b_1 = b_2$ and $b_3 = b_4$)
- We have to rewrite these q constraints in matrix form: $Rb = r$, with R a $q \times k$ matrix
- Ex. with $b = (b_0, b_1, b_2, b_3, b_4)'$: $R = \begin{pmatrix} 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 & -1 \end{pmatrix}$, $r = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

The test statistic is the following:

$$f = \frac{1}{q}(R\hat{b} - r)'(\hat{\sigma}^2 R(X'X)^{-1}R')^{-1}(R\hat{b} - r) \hookrightarrow F_{(q,N-k)}$$

How to implement this? (1)

- In fact, all this amounts to testing whether the unconstrained model is significantly better than the constrained model
- Let's call R^2_{nc} the R^2 of the non constrained model, and R^2_c the R^2 of the constrained model
- It can be shown that:

$$f = \frac{(R^2_{nc} - R^2_c)}{1 - R^2_{nc}} \cdot \frac{N-k}{q} \hookrightarrow F_{(q,N-k)}$$

With N: sample size, k: no. of parameters of the non constrained model (constant included), q: no. of constraints. Valid only if both models have the same dependent variable.

How to implement this? (2)

- Call SSR the sum of the squared residuals
- df: the residual degrees of freedom of each model (N minus its number of parameters)
- c: constrained, nc: non constrained models
- It can be shown that:

$$f = \frac{(SSR_c - SSR_{nc})}{SSR_{nc}} \cdot \frac{df_{nc}}{df_c - df_{nc}} \hookrightarrow F_{(df_c - df_{nc}, df_{nc})}$$

How to select variables

- Choose to use variables in levels, logs or growth rates using economic theory, variable plots, tests or common sense
- Detect perfect multicollinearity and remove it
- Then, 3 options:
- Backward elimination: estimate the model with all the variables and remove those that are not significant (T then F tests)
- Forward selection: begin with only one explanatory variable, the one that is the most correlated with the dependent variable, then introduce the other ones one by one according to their partial correlation coefficient with y, while they are significant
- Stepwise regression: same process as forward, and at each step variables that cease to be significant are removed
- No automatic procedure can replace common sense

Outline

1. Tests
2. Size, p-value, forecasting
3. Usual tests: T and F
4. Testing for structural change
   - Purpose
   - Computation
   - Non constant variances
   - Various issues

The Chow test

- It could be relevant to test whether parameters are the same on 2 subsamples of the data
- Ex: are consumer behaviors the same before and after the German reunification? Idea: test the temporal stability of the regression
- Ex: are the parameters of a wage equation the same for men and women? Idea: test homogeneity of behaviors
- Since we are testing a set of parameters simultaneously, we use an F-test
- The constrained model is the one where parameters are forced to be the same on the two subsamples
- The non constrained model is the one where parameters are allowed to differ across the two subsamples
- The test will tell us if we can or cannot reject homogeneity of behaviors

1st method

Let's call SSR the sum of squared residuals

- SSR_c: of the constrained model (estimated on the whole sample)
- SSR_1, SSR_2: of the model estimated on the 1st and the 2nd subsample respectively
- The non constrained model's SSR is then $SSR_{nc} = SSR_1 + SSR_2$

$$F = \frac{(SSR_c - SSR_{nc})/k}{SSR_{nc}/(N-2k)} \hookrightarrow F(k, N-2k)$$

- Rmk 1: this method requires the estimation of 3 models
- Rmk 2: this is the general F formula (with SSR or R^2), with $q = k$ constraints

2nd method

We can run the Chow test by estimating one single model, using interacted variables

- Initial model, for each observation i: $y_i = b_0 + \sum_{j=1}^{k-1} b_j x_{j,i} + u_i$
- Create a dummy I_i equal to 1 if observation i belongs to the 1st subsample, and 0 otherwise
- For each explanatory variable, create $z_{j,i} = x_{j,i} \times I_i$ (equal to $x_{j,i}$ in the 1st subsample, 0 otherwise)
- Run the "augmented" regression: $y_i = b_0 + \sum_{j=1}^{k-1} b_j x_{j,i} + a_0 I_i + \sum_{j=1}^{k-1} a_j z_{j,i} + u_i$
- Parameters are the same across the 2 subsamples if adding these new variables doesn't improve the initial model
- The Chow test thus amounts to an F-test of $H_0 : a_0 = a_1 = \cdots = a_{k-1} = 0$ (k constraints)

Two remarks

- The 2 methods presented here are totally equivalent
- Implicitly, this test assumes that the variance of the error term is the same in the two subsamples, which is not always the case (see below)
- If the second subsample is too small (fewer observations than parameters), we can't estimate the model on it. In that case, we use the Chow *predictive* test (see below)

Variance of error terms not constant

- Parameters could differ across 2 subperiods, and so could the variance of the error terms
- If it is the case, it can be shown that the actual size of the test differs from the nominal α: we may reject the hypothesis of homogeneity of behaviors too often
- The Chow test needs to be adjusted in that case
- The fact that error terms can have a non constant variance is called *heteroskedasticity*, and will be studied later

Test of equality of variances

- We deal with subsamples 1 and 2, calling T_1 and T_2 their sizes and k the number of parameters in the model
- Assume $u_1 \hookrightarrow N(0, \sigma_1^2)$ and $u_2 \hookrightarrow N(0, \sigma_2^2)$
- We assume a constant variance within each subsample, while σ_1 and σ_2 are not necessarily equal
- We thus have: $\frac{\hat{u}_1' \hat{u}_1}{\sigma_1^2} \hookrightarrow \chi^2_{T_1-k}$ and $\frac{\hat{u}_2' \hat{u}_2}{\sigma_2^2} \hookrightarrow \chi^2_{T_2-k}$
- We know that the ratio of two independent χ^2, each divided by its degrees of freedom, gives a Fisher distribution
- Under the null hypothesis $H_0 : \sigma_1 = \sigma_2$, this ratio is thus:

$$C = \frac{\hat{u}_1' \hat{u}_1}{T_1 - k} / \frac{\hat{u}_2' \hat{u}_2}{T_2 - k} = \frac{\hat{\sigma}_1^2}{\hat{\sigma}_2^2} \hookrightarrow F(T_1 - k, T_2 - k)$$

Remark

- Subsamples 1 and 2 are apparently interchangeable
- In practice, we can compute two test statistics: one > 1 and the other < 1, because one variance will always be greater than the other
- We commonly use the statistic that is > 1, and compare it with the upper quantile of the Fisher distribution

Consequence on the Chow test

- Assume we proved that $\sigma_1 \neq \sigma_2$
- Let's use the ratio $w = \frac{\sigma_1}{\sigma_2}$
- If we multiply the second sample's model by w, we make the variances of the error terms the same for the two models
- Ex: $y_2 = a_2 + b_2 x_2 + u_2$ with $V(u_2) = \sigma_2^2$ becomes:
- $wy_2 = wa_2 + wb_2 x_2 + wu_2$, with the variance of the new error term: $V(wu_2) = (\frac{\sigma_1}{\sigma_2})^2 \sigma_2^2 = \sigma_1^2$
- Since we don't know the "true" w because we don't know the "true" σ_1 and σ_2, we replace them by their estimates, computed when running the variance equality test
- I.e.: $\hat{w} = \frac{\hat{\sigma}_1}{\hat{\sigma}_2}$, and we then run the usual Chow test on the reweighted data

What if the second subsample is too small

Ex: the second subperiod is too short ($T_2 < k$). We can still run a test, with the following adjustments:

- The model is estimated on the first subperiod ($y_1 = X_1 b_1 + u_1$) to get parameters $\hat{b}_1$
- Comparing the prediction $X_2 \hat{b}_1$ and the observation y_2 is a way of testing whether model 1 is still valid on period 2
- More precisely, we test whether the prediction error has expectation zero
- Calling SSR_c the sum of squared residuals of the constrained model (estimated on both periods), SSR_1 that of model 1, and T_1 and T_2 the sizes of samples 1 and 2, the following statistic can be used to run a Chow (predictive) test:

$$\frac{(SSR_c - SSR_1)/T_2}{SSR_1/(T_1 - k)} \hookrightarrow F(T_2, T_1 - k)$$

Remark

- The idea of the Chow test is to compare the constrained vs. the non constrained model
- Estimating a model with more parameters than observations gives non-unique parameters
- If $T_2 \leq k$, the model can fit subsample 2 perfectly, so all its residuals are zero
- That's why we end up with a formula close to the usual Chow test, where SSR_2 would be zero

Choosing a breakpoint in a time series

Unless we know a specific event happened, finding a breakpoint is not obvious

- This is the purpose of the CUSUM (cumulative sum) test
- Method: analyze the prediction of y_t computed with the parameters of the model estimated on the first $t-1$ periods
- The test can begin at date $t = k+1$ at the earliest
- If the regression is stable over time, then the expectation of the forecast error is 0
- We plot the cumulative sum of these so-called *recursive residuals* against time: if the model is stable over time, it must stay within some confidence bounds

For more details, see Cadoret et al., chapter 3.